Scanning with Qt-DAB-7.3.1

Jan van Katwijk, Lazy Chair Computing
The Netherlands

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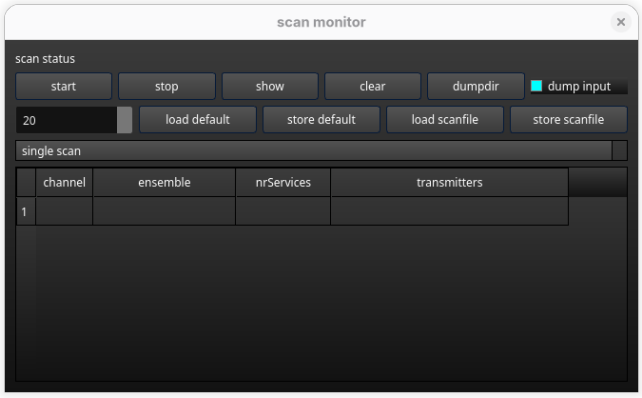
1 Introduction

Qt-DAB provides *scanning* of the DAB band and showing (and storing) the results. During scanning, the TII data is collected as in non scanning modes, and the resulting transmitters can be made visible on the map during or after the scan. Qt-DAB supports *three* modes for scanning:

1. *Scanning single*, the current band is scanned from the first channel up to the last in the band. Single scanning is the most used mode for scanning, a single scan is performed over the band, and on a separate window an overview of the contents of the channels with DAB data is presented.
2. *Scanning the current band until* a channel is detected with recognizable DAB data; Scanning is started in the currently selected channel and continues until a channel is encountered with DAB data. If no such channel is detected, scanning continues until explicitly stopped;
3. *Scanning continuously*, starting at the first channel in the band, scanning the band continues until the user explicitly stops the scan. For each channel where DAB data is detected, a small (one or two lines) summary is shown on what was encountered in that channel.

2 The scan controller

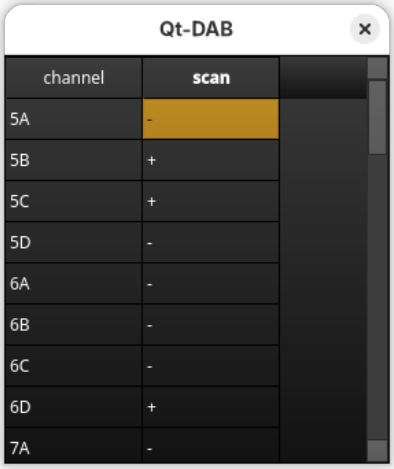
All scanning activities are controlled by the so called *scan controller*. Visibility of this scan controller is controlled by the button labeled *scan* on the main window.



The topline of the scan controller shows the status, when scanning it tells the channel being visited.

The *scan mode* is set on the forth line. As mentioned above, the mode is either *single scan*, i.e case 1 above, *until data*, i.e. case 2 above or *continuous*, i.e. case 3 above.

It is often quite obvious that some channels are better skipped when scanning, it is known that they do not contain DAB data. Qt-DAB provides for that purpose so-called *scan files*. A scan file is a file containing a list of channels, where each channel is marked whether it should be skipped or not.



Two types of scan file are provided for:

1. a *default* one, the data of which is just recorded in the *.ini* file and
2. *named ones*, the data of which is stored into a file with a user defined name.

The *third* line contains (from left to right):

1. a time indicator, telling the number of seconds to stay interpreting data in a selected channel (default 20 seconds);
2. the *load default* button, when touched the *default* scan list is shown (and can be manipulated);
3. the *store default* button, when touched the *default* scan list is stored (as the name suggests);
4. the *load scanfile* button, when touched a file selection menu appears and *an existing* file can be selected;
5. the *store scanfile* button, when touched the content of the current scan list is stored in a user defined scan file.

Note that the regular way to create a user defined scan file is by:

1. loading the default scan file;
2. modifying it to your wishes, and
3. storing it as a user defined scan file.

The scanlist itself is just a table with channel names, each with an indicator telling whether or not the channel should be visited when scanning.

The *second line* of the scan controller contains the actual control buttons:

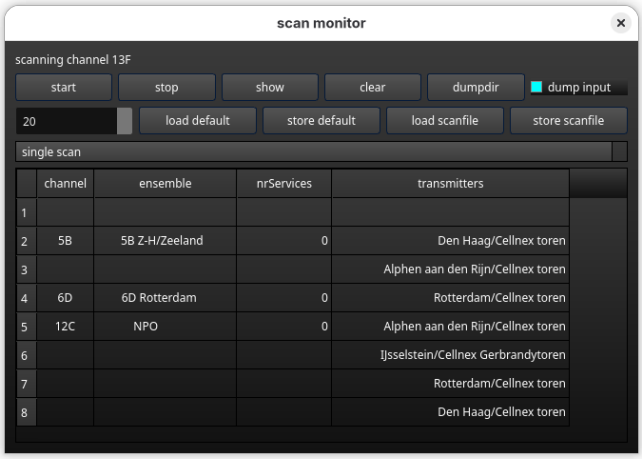
1. the *start* button starts the actual scanning, and
2. the *stop* button stops the scanning (obviously if scanning is on);
3. the *show* button shows the currently selected scanlist;
4. the *clear* button clears the scan list, i.e. all channels are visited;
5. the *dumpdir* button allows selection of a directory (folder) to storing the scan list(s);
6. the *dump input* button, when touched, causes the raw sample input from each channel to be written into a file, which is then written in ".csv" format.

3 Scan output

3.1 Single Scan

The output of the scanner is split up into two parts:

1. in the scan control window, the channel where data was detected, together with the ensemble name, the number of services detected, and the transmitters detected with that channel is shown;
2. in a separate window, the details per channel are shown.



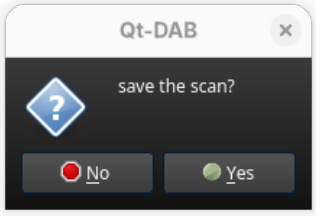
Unfortunately, the number of services detected in the channels is not computed correctly, this will be fixed in a next version.

Qt-DAB

	current ensemble	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1															
2	ensemble	channelName	frequency (KHz)	Eid	time	SNR	nr services								
3	...														
4	5B Z-H/Zeeland	5B	Netherlands	176640000	805b	14 Oct 2026 ...	SNR 10	14	108, 12%						
5	...														
6		(15-7)	Den Haag/...	11.0km	313.5 °	1.0m	116.0m	...							
7		(15-13)	Alphen aan de...	19.9km	45.9 °	0.0m	124.0m	...							
8	audio services														
9		serviceName	short name	serviceId	subChannel	start address ...	length (CU's)	protection	code rate	bitrate	dab type	language	program type	fm freq	(opt. country)
10		RADIONL	RADIONL	8421	6	252	36	EEP 3-A	1/2	48	DAB+	unknown	Pop Music		
11		Amor FM	Amor FM	845a	8	360	72	EEP 3-A	1/2	96	DAB+	unknown	Varied		
12		Radio Rijnmond	Rijnmond	841a	4	180	72	EEP 3-A	1/2	96	DAB+	unknown	Current Affairs		
13		Omroep ...	Omr Zlnd	841b	1	36	72	EEP 3-A	1/2	96	DAB+	unknown	Current Affairs	87900	
14		Tukker FM	TukkerFM	8457	11	486	36	EEP 3-A	1/2	48	DAB+	unknown	Pop Music		
15		ARROW ...	BLUESBOX	8ab1	2	108	72	EEP 3-A	1/2	96	DAB+	unknown	Rock Music		
16		THE GROOVE	GROOVE	84ae	15	684	36	EEP 3-A	1/2	48	DAB+	unknown	Pop Music		
17		KISS FM	KISS FM	840d	13	576	54	EEP 3-A	1/2	72	DAB+	unknown	Pop Music		
18		GirlPower	GrlPwr	8dfd	14	630	54	EEP 3-A	1/2	72	DAB+	unknown	Pop Music		
19		GLXY.RADIO	GLXY	8dad	12	522	54	EEP 3-A	1/2	72	DAB+	unknown	National Music		
20		Sunrise FM	Sunrise	8daf	16	720	36	EEP 3-A	1/2	48	DAB+	unknown	Varied		
21		gpr ...	gprSPORT	8998	0	0	36	EEP 3-A	1/2	48	DAB+	unknown	Sport		
22		Hofstad Radio	Hofstad	8e02	9	432	54	EEP 3-A	1/2	72	DAB+	unknown	Pop Music		

As shown, the details of the ensemble detected in the channels are presented in the same format as used with the content dump of the current ensemble.

At the end of the scan, one is asked whether or not contents description is to be saved into a file in ".csv" format.

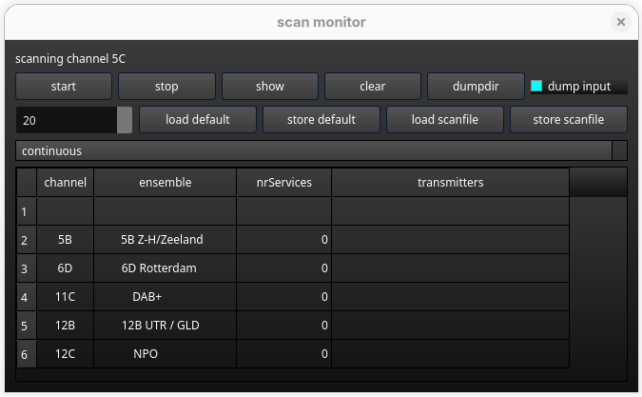


3.2 Scan to output

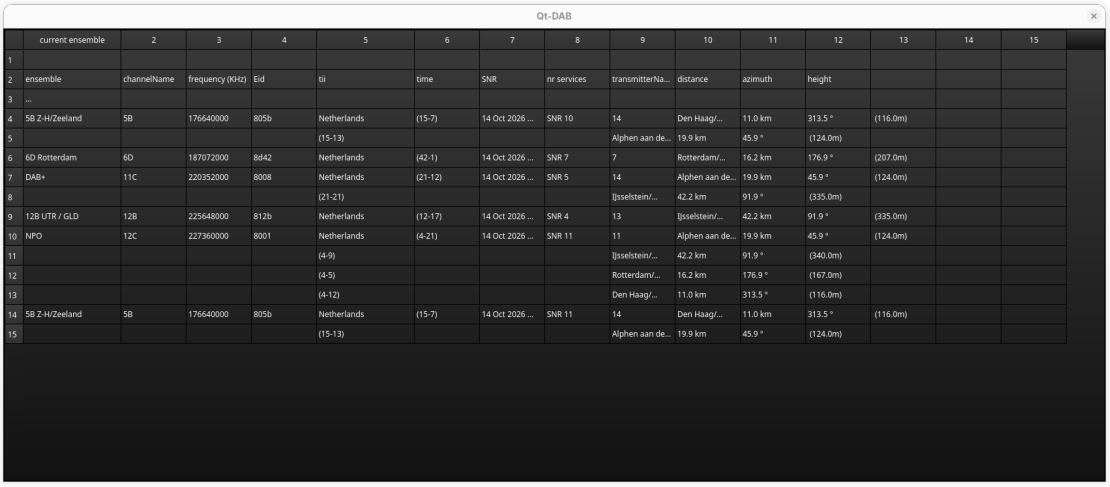
Scan to output does not show output.

3.3 Scan continuously

When scanning continuously, the *per channel* data is somewhat limited as no service names are shown. The scan controller shows just the names of the ensembles detected



A separate window gives per channel information in one or a few lines. Note that the headings do not always match the content of the column, this document describes the current version. It will be fixed in a next version of Qt-DAB.



Since the scanning continues, the common way of stopping the scan is by touching the *stop* button as mentioned above. After having touched that button such that the scan stops, one is asked whether or not to save the contents of the separate window in a ".csv" type file.

